

APEXPLANET

**CYBERSECURITY & ETHICAL HACKING
INTERNSHIP PROGRAM**

TASK – 2

NETWORK SECURITY & SCANNING

Practical Task Report

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Task	Task – 2
Current Scope	Reconnaissance & Port / Service Scanning

Authorized Laboratory Target: Metasploitable 2 • 192.168.56.104

Table of Contents

- 1. Introduction
- 2. Task Objective
- 3. Laboratory Environment and Scope
- 4. Reconnaissance
 - 4.1 Passive Reconnaissance
 - 4.1.1 WHOIS
 - 4.1.2 Nslookup
 - 4.1.3 Google Dorking
 - 4.1.4 Shodan
 - 4.2 Active Reconnaissance
 - 4.2.1 Ping Sweep
 - 4.2.2 Banner Grabbing
- 5. Port & Service Scanning
 - 5.1 TCP SYN Scan
 - 5.2 UDP Scan
 - 5.3 Service / Version Detection
 - 5.4 OS Detection
 - 5.5 Consolidated Nmap Scan Report

1. Introduction

This report documents the practical activities completed under Task 2 – Network Security & Scanning of the Cybersecurity & Ethical Hacking Internship Program. The work covered reconnaissance, network host discovery, port enumeration, service identification, and operating system detection using a controlled laboratory environment.

The reconnaissance phase included both passive and active techniques. Passive reconnaissance was demonstrated using WHOIS, Nslookup, Google search operators, and Shodan to examine publicly available information without directly probing the target infrastructure. Active reconnaissance was performed within the authorized Host-Only laboratory network using Nmap host discovery and service/version detection against the Metasploitable 2 virtual machine.

The port and service scanning phase further examined the authorized target using TCP SYN scanning, UDP scanning, service/version detection, and operating system detection. Together, these activities establish the target's exposed network surface and provide a technical baseline for the subsequent vulnerability assessment and packet-analysis activities of Task 2.

2. Task Objective

To learn and demonstrate reconnaissance, network scanning, service identification, and network traffic analysis techniques in an authorized cybersecurity laboratory environment.

The completed work in this draft focuses on reconnaissance and port/service scanning. The remaining Task 2 activities will be added to the report as they are completed.

3. Laboratory Environment and Scope

All active scanning activities were restricted to the authorized Host-Only laboratory network and the Metasploitable 2 virtual machine used for practice.

Attacker / Analyst VM	Kali Linux
Laboratory Network	192.168.56.0/24
Authorized Target	Metasploitable 2 – 192.168.56.104
Scanning Tool	Nmap 7.98

3.1 Initial Connectivity Verification

Initial network configuration and connectivity were verified between the Kali Linux system and the Metasploitable laboratory machine before active reconnaissance was performed.

```
Session Actions Edit View Help
(kali@kali)~$ ip addr
1: lo:<LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0:<BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:b4:a1:05 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 86371sec preferred_lft 86371sec
   inet6 fd17:625c:f037:2:ac18:4f09:29f1:b746 scope global dynamic noprefixroute
       valid_lft 86373sec preferred_lft 14373sec
   inet6 fe80::e652:339d:e511:a764 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: eth1:<BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:28:22:d5 brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.104/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
       valid_lft 571sec preferred_lft 571sec
   inet6 fe80::35fc:253b:7fec:164 scope link noprefixroute
       valid_lft forever preferred_lft forever
(kali@kali)~$
```

Figure 1. Kali Linux network interface configuration.

```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:4b:af:2c
          inet addr:192.168.56.104  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe4b:af2c/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:3 errors:0 dropped:0 overruns:0 frame:0
          TX packets:29 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1313 (1.2 KB)  TX bytes:3638 (3.5 KB)
          Base address:0xd020  Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:92 errors:0 dropped:0 overruns:0 frame:0
          TX packets:92 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19393 (18.9 KB)  TX bytes:19393 (18.9 KB)

msfadmin@metasploitable:~$
```

Figure 2. Metasploitable 2 network interface configuration.

```
(kali@kali)~$ ping -c 4 192.168.56.104
PING 192.168.56.104 (192.168.56.104) 56(84) bytes of data.
64 bytes from 192.168.56.104: icmp_seq=1 ttl=64 time=13.0 ms
64 bytes from 192.168.56.104: icmp_seq=2 ttl=64 time=2.90 ms
64 bytes from 192.168.56.104: icmp_seq=3 ttl=64 time=6.36 ms
64 bytes from 192.168.56.104: icmp_seq=4 ttl=64 time=2.41 ms

--- 192.168.56.104 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3094ms
rtt min/avg/max/mdev = 2.405/6.162/12.978/4.220 ms
```

Figure 3. Initial ICMP connectivity test to 192.168.56.104.

4. Reconnaissance

4.1 Passive Reconnaissance

4.1.1 WHOIS

Objective

To perform passive reconnaissance and collect publicly available domain registration information using WHOIS.

Tool Used

WHOIS

Target

microsoft.com

Command / Query

whois microsoft.com

Procedure

1. Opened the Kali Linux terminal.
2. Executed the WHOIS query for microsoft.com.
3. Examined the returned registrar, registration-date, domain-status, and name-server information.

Result

The WHOIS query returned publicly available registration information, including registrar details, domain dates, domain status, and name-server information.

```
Session Actions Edit View Help
$ whois microsoft.com
Domain Name: MICROSOFT.COM
Registry Domain ID: 2724960_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2026-01-29T18:46:57Z
Creation Date: 1991-05-02T04:00:00Z
Registry Expiry Date: 2027-05-03T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1-39.AZURE-DNS.COM
Name Server: NS2-39.AZURE-DNS.NET
Name Server: NS3-39.AZURE-DNS.ORG
Name Server: NS4-39.AZURE-DNS.INFO
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-08-15T11:42:12Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.
```

Figure 4. WHOIS query results for microsoft.com.

Analysis

WHOIS can provide useful publicly available information during reconnaissance. Such information can help an analyst understand domain registration infrastructure and identify associated name servers and other domain-related details.

Conclusion

WHOIS reconnaissance was successfully demonstrated using a public domain.

4.1.2 Nslookup

Objective

To perform DNS reconnaissance using the Nslookup utility and identify the IP address associated with a given domain name.

Tool Used

Nslookup

Target

microsoft.com

Command / Query

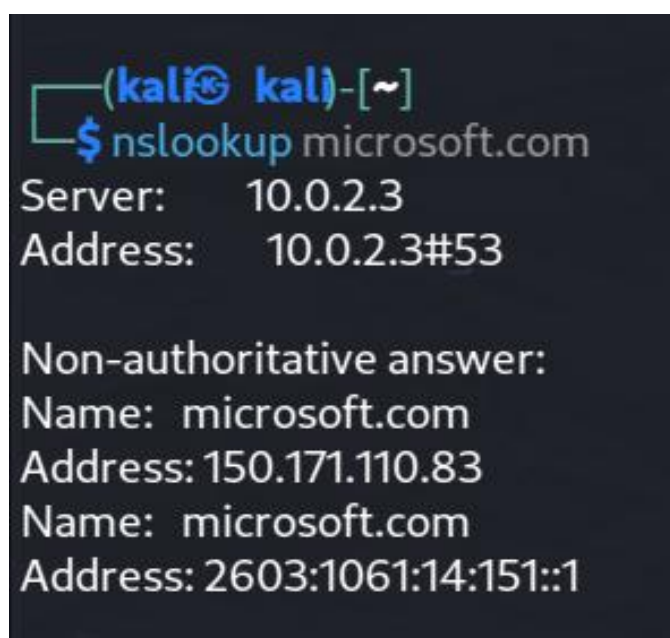
nslookup microsoft.com

Procedure

4. Opened the Kali Linux terminal.
5. Executed the nslookup microsoft.com command.
6. The command queried the configured DNS server for DNS information associated with microsoft.com.
7. Observed the returned IPv4 and IPv6 addresses.

Result

The DNS query successfully resolved microsoft.com. The configured DNS server was 10.0.2.3. The returned IPv4 address was 150.171.110.83 and the returned IPv6 address was 2603:1061:14:151::1. The response was identified as non-authoritative.



```
(kali kali)-[~]  
$ nslookup microsoft.com  
Server:      10.0.2.3  
Address:     10.0.2.3#53  
  
Non-authoritative answer:  
Name: microsoft.com  
Address: 150.171.110.83  
Name: microsoft.com  
Address: 2603:1061:14:151::1
```

Figure 5. Nslookup result for microsoft.com.

Analysis

The command successfully provided DNS resolution information for the queried domain. DNS reconnaissance can help identify IP addresses associated with a domain during the initial reconnaissance phase.

Conclusion

Nslookup reconnaissance was successfully performed and the DNS information returned by the resolver was recorded.

4.1.3 Google Dorking

Objective

To perform passive reconnaissance using Google search operators and identify publicly indexed information associated with a specified domain.

Tool Used

Google Search

Target

example.com

Command / Query

site:example.com

Procedure

8. Opened Google Search in the Kali Linux browser.
9. Entered the search operator site:example.com.
10. Examined the search results returned by Google.
11. Identified publicly indexed pages associated with the specified domain.

Result

The search returned results associated with example.com, including the publicly accessible Example Domain webpage. The site: operator restricted the search results to pages indexed under the specified domain.

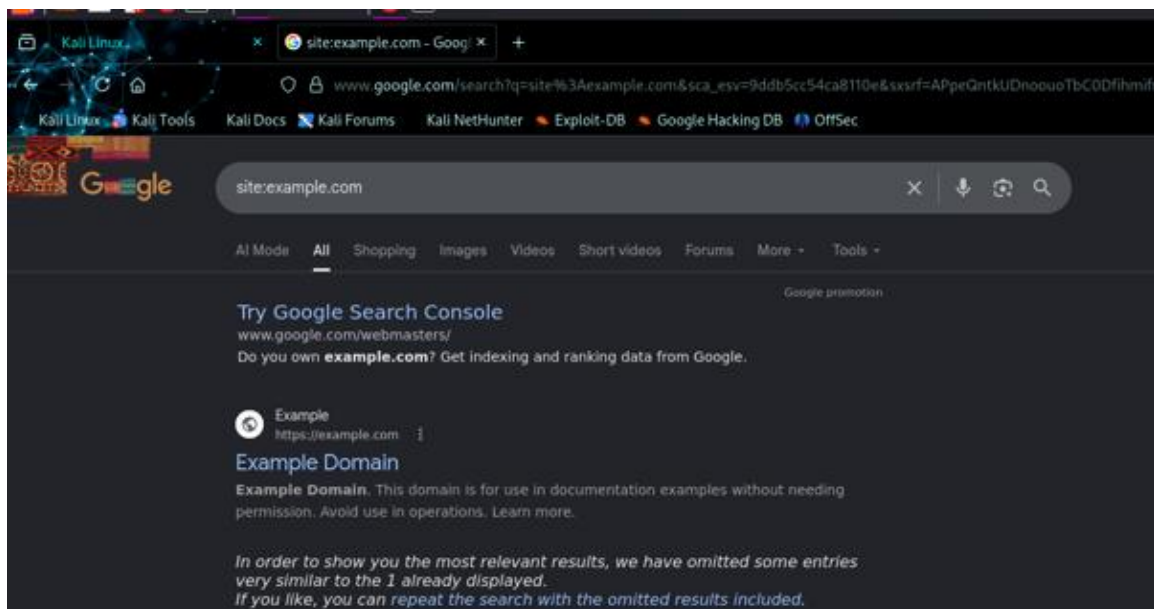


Figure 6. Google Dorking using the site: search operator.

Analysis

The site: search operator can be used during passive reconnaissance to identify webpages indexed for a particular domain. This helps an analyst understand the publicly visible web footprint without directly interacting with the target infrastructure.

Conclusion

Google Dorking was successfully demonstrated using the site: operator.

4.1.4 Shodan

Objective

To perform passive reconnaissance using Shodan and examine publicly indexed information about internet-connected services and systems.

Tool Used

Shodan

Command / Query

apache

Procedure

12. Opened the Shodan search engine in the browser.
13. Entered apache in the search field.
14. Examined the results returned by Shodan.
15. Observed publicly indexed information related to systems running or exposing Apache services.

Result

The search returned numerous Shodan-indexed results associated with Apache services. The displayed information included IP addresses, hostnames, organizations, geographic locations, HTTP response information, and detected server software. One displayed service identified Apache Tomcat/4.0.3.

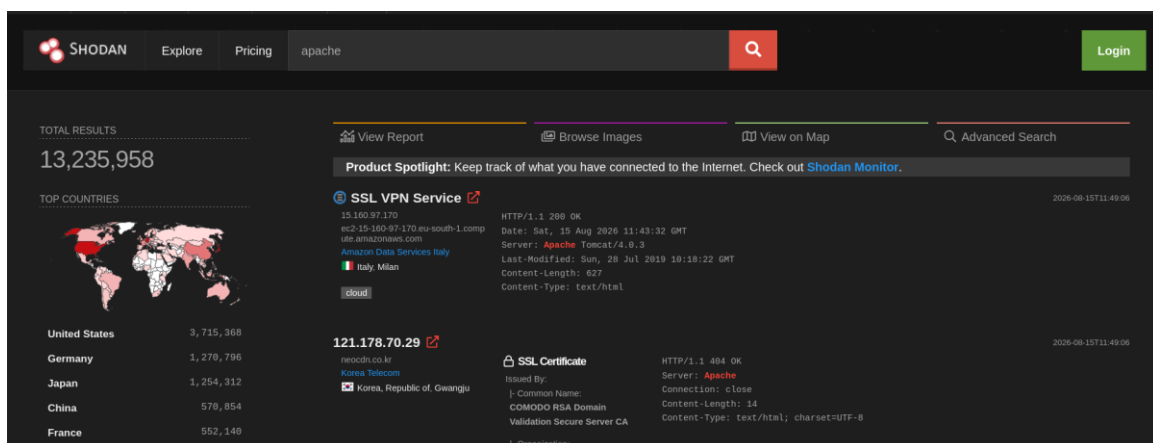


Figure 7. Shodan search results for Apache services.

Analysis

Shodan provides a search interface for information that it has already collected and indexed from internet-connected systems. During passive reconnaissance, this can help analysts understand publicly exposed technology and service footprints without directly scanning the displayed systems.

Conclusion

Shodan reconnaissance was successfully performed and publicly indexed Apache-related service information was observed.

4.2 Active Reconnaissance

4.2.1 Ping Sweep

Objective

To identify active hosts within the authorized Host-Only laboratory network using Nmap host discovery.

Tool Used

Nmap 7.98

Target

192.168.56.0/24

Command / Query

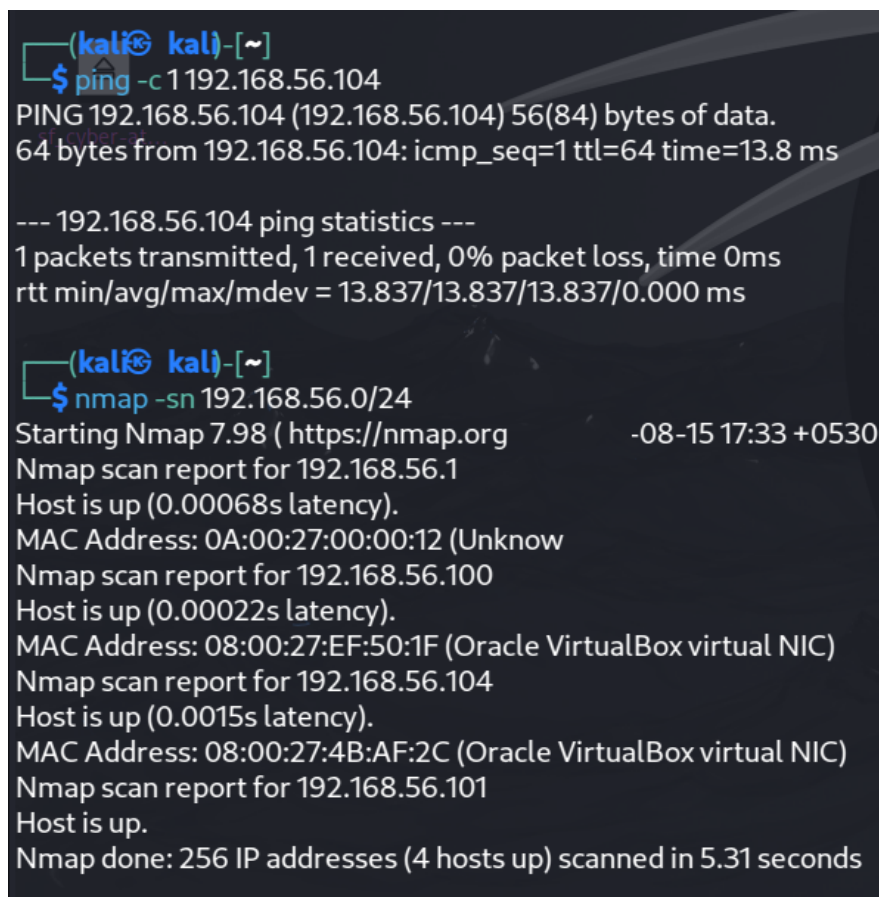
```
nmap -sn 192.168.56.0/24
```

Procedure

16. Opened the Kali Linux terminal.
17. Identified the Host-Only laboratory network as 192.168.56.0/24.
18. Executed the Nmap host-discovery command.
19. Examined the response to identify active hosts.

Result

The scan covered 256 IP addresses and identified four active hosts: 192.168.56.1, 192.168.56.100, 192.168.56.101, and 192.168.56.104. The Metasploitable target at 192.168.56.104 was confirmed to be active.



```
(kali㉿ kali)-[~]
$ ping -c 1 192.168.56.104
PING 192.168.56.104 (192.168.56.104) 56(84) bytes of data:
64 bytes from 192.168.56.104: icmp_seq=1 ttl=64 time=13.8 ms

--- 192.168.56.104 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 13.837/13.837/13.837/0.000 ms

(kali㉿ kali)-[~]
$ nmap -sn 192.168.56.0/24
Starting Nmap 7.98 ( https://nmap.org ) -08-15 17:33 +0530
Nmap scan report for 192.168.56.1
Host is up (0.00068s latency).
MAC Address: 0A:00:27:00:00:12 (Unknow
Nmap scan report for 192.168.56.100
Host is up (0.00022s latency).
MAC Address: 08:00:27:EF:50:1F (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.104
Host is up (0.0015s latency).
MAC Address: 08:00:27:4B:AF:2C (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.101
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 5.31 seconds
```

Figure 8. Nmap ping sweep / host discovery results.

Analysis

Host discovery successfully identified active systems within the authorized laboratory network. This establishes which hosts are available for subsequent service and port enumeration.

Conclusion

Four active hosts were identified, including the authorized Metasploitable target.

4.2.2 Banner Grabbing

Objective

To identify open services and determine service names and software versions running on the authorized Metasploitable target using Nmap service detection.

Tool Used

Nmap 7.98

Target

192.168.56.104 – Metasploitable

Command / Query

nmap -sV 192.168.56.104

Procedure

20. Opened the Kali Linux terminal.
21. Specified the Metasploitable system as the authorized laboratory target.
22. Executed Nmap service-version detection.
23. Examined the identified services and software versions.

Result

The scan identified multiple open TCP services, including FTP (vsftpd 2.3.4), SSH (OpenSSH 4.7p1), Telnet, SMTP, DNS, HTTP (Apache httpd 2.2.8), Samba, MySQL 5.0.51a, PostgreSQL, VNC, AJP13, and Apache Tomcat.

```
Session Actions Edit View Help
└─$ nmap -sV 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-15 17:35 +0530
Nmap scan report for 192.168.56.104
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        netkit-rshd
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:4B:AF:2C (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 13.43 seconds
```

Figure 9. Nmap service and version detection results.

Analysis

The scan revealed a large number of exposed services and identified software versions where available. This information provides useful context for subsequent vulnerability assessment because software versions can be compared with known vulnerability information.

Conclusion

Banner grabbing was successfully performed and multiple open services and corresponding software information were identified.

5. Port & Service Scanning

5.1 TCP SYN Scan

Objective

To identify open TCP ports on the authorized Metasploitable target using an Nmap TCP SYN scan.

Tool Used

Nmap 7.98

Target

192.168.56.104 – Metasploitable

Command / Query

`nmap -sS 192.168.56.104`

Procedure

24. Opened the Kali Linux terminal.
25. Specified the Metasploitable system as the scan target.
26. Executed the Nmap TCP SYN scan using `-sS`.
27. Examined the returned port states and identified open TCP ports.

Result

The scan identified 23 open TCP ports, including 21, 22, 23, 25, 53, 80, 111, 139, 445, 512, 513, 514, 1099, 1524, 2049, 2121, 3306, 5432, 5900, 6000, 6667, 8009, and 8180. Nmap reported 977 closed TCP ports.

```
(kali) kali ~  
$ nmap -sS 192.168.56.104  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-15 17:40 +0530  
Nmap scan report for 192.168.56.104  
Host is up (0.0014s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 08:00:27:4B:AF:2C (Oracle VirtualBox virtual NIC)
```

Figure 10. TCP SYN scan results for the Metasploitable target.

Conclusion

The TCP SYN scan was successfully performed and multiple open TCP services were identified on the authorized Metasploitable system.

5.2 UDP Scan

Objective

To identify open and potentially accessible UDP services on the authorized Metasploitable target using Nmap UDP scanning.

Tool Used

Nmap 7.98

Target

192.168.56.104 – Metasploitable

Command / Query

```
nmap -sU --top-ports 20 192.168.56.104
```

Procedure

28. Opened the Kali Linux terminal.
29. Specified the authorized Metasploitable system as the target.
30. Performed a UDP scan using -sU.
31. Limited the scan to the top 20 commonly used UDP ports for efficient execution.
32. Examined the resulting port states and services.

Result

The scan identified 53/udp (domain) and 137/udp (netbios-ns) as open. Ports 68/udp, 69/udp, and 138/udp were reported as open|filtered. The scan completed in 16.10 seconds.



```
(kali) kali-[-~]
$ nmap -sU --top-ports 20 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-15 17:59 +0530
Nmap scan report for 192.168.56.104
Host is up (0.0015s latency).

PORT      STATE      SERVICE
53/udp    open       domain
67/udp    closed     dhcpc
68/udp    open|filtered dhcpc
69/udp    open|filtered tftp
123/udp   closed     ntp
135/udp   closed     msrpc
137/udp    open       netbios-ns
138/udp   open|filtered netbios-dgm
139/udp   closed     netbios-ssn
161/udp   closed     snmp
162/udp   closed     snmptrap
445/udp   closed     microsoft-ds
500/udp   closed     isakmp
514/udp   closed     syslog
520/udp   closed     route
631/udp   closed     ipp
1434/udp  closed     ms-sql-m
1900/udp  closed     upnp
4500/udp  closed     nat-t-ike
49152/udp closed     unknown
MAC Address: 08:00:27:4B:AF:2C (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 16.10 seconds
```

Figure 11. UDP scan results for the Metasploitable target.

Analysis

UDP scanning complements TCP scanning because UDP-based services can expose a different portion of a system's network attack surface. The open|filtered state indicates that Nmap could not definitively distinguish an open port from a filtered port based on the available UDP response.

Conclusion

UDP scanning was successfully performed and multiple UDP service states were identified.

5.3 Service / Version Detection

Objective

To identify network services and associated software versions running on the authorized Metasploitable target.

Tool Used

Nmap 7.98

Target

192.168.56.104 – Metasploitable

Command / Query

nmap -sV 192.168.56.104

Procedure

33. Executed Nmap service-version detection against the authorized target.
34. Examined the service names and version information returned by Nmap.
35. Recorded the identified technologies for later analysis.

Result

The service-version scan identified multiple services and software versions, including vsftpd 2.3.4 on FTP, OpenSSH 4.7p1, Apache httpd 2.2.8, ISC BIND 9.4.2, Samba 3.X–4.X, MySQL 5.0.51a, PostgreSQL 8.3.0–8.3.7, VNC, Apache Jserv, and Apache Tomcat/Coyote JSP engine 1.1.

```
Session Actions Edit View Help
└─$ nmap -sV 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-15 17:35 +0530
Nmap scan report for 192.168.56.104
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        Netkit rshd
514/tcp   open  shell        Netkit rshd
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:4B:AF:2C (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 13.43 seconds
```

Figure 12. Service/version detection evidence reused from the banner-grabbing scan.

Analysis

The service-version information provides a more detailed view of the exposed attack surface than port state alone. The same evidence was used during banner grabbing and is retained here because service/version detection is an explicit requirement of the scanning phase.

Conclusion

Service and version detection was successfully completed using Nmap.

5.4 OS Detection

Objective

To identify the operating system and system characteristics of the authorized Metasploitable target using Nmap OS detection.

Tool Used

Nmap 7.98

Target

192.168.56.104 – Metasploitable

Command / Query

nmap -O 192.168.56.104

Procedure

36. Specified the authorized Metasploitable system as the target.
37. Executed Nmap with the -O option.
38. Examined the OS fingerprint and host characteristics reported by Nmap.

Result

Nmap identified the target as a general-purpose Linux system. The reported running family was Linux 2.6.X, with OS details estimated as Linux 2.6.9–2.6.33. Network distance was reported as 1 hop.

```
Session Actions Edit View Help
└─$ nmap -O 192.168.56.104
Starting Nmap 7.98 ( https://nmap.org )      8-15 18:05 +0530
Nmap scan report for 192.168.56.104
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:4B:AF:2C (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.33 seconds
```

Figure 13. Nmap operating system detection results.

Analysis

The result is an Nmap fingerprint/estimate rather than absolute confirmation of the exact kernel version. OS identification is useful because operating-system characteristics can help guide later vulnerability and configuration analysis.

Conclusion

OS detection was successfully performed and the target was identified as a Linux-based general-purpose system.

5.5 Consolidated Nmap Scan Report

The consolidated scan report will summarize the TCP, UDP, service/version, and OS-detection findings obtained from the authorized Metasploitable target. This section can be expanded with the final scan analysis and vulnerability correlations after the remaining Task 2 activities are completed.

Category	Command	Key Finding	Evidence
TCP	<code>nmap -sS 192.168.56.104</code>	23 open TCP ports; 977 closed	Figure 10
UDP	<code>nmap -sU --top-ports 20 192.168.56.104</code>	53/udp and 137/udp open; several open filtered	Figure 11
Services	<code>nmap -sV 192.168.56.104</code>	Multiple services and versions identified	Figure 9 / 12
OS	<code>nmap -O 192.168.56.104</code>	Linux 2.6.x fingerprint; 1 hop	Figure 13

Note: The remaining Task 2 sections—vulnerability scanning, packet analysis, firewall basics, and the final deliverables—should be added after their practical execution so that the report remains evidence-based and consistent with the actual lab results.